

## **Lab Work**

### **Simple Java Programs:**

- 1) Write a program to perform basic arithmetic operations (addition, subtraction, multiplication, division) using variables and operators.
- 2) Create a program to convert temperature between Celsius and Fahrenheit using variables and arithmetic operators.
- 3) Write a program to check if a given year is a leap year or not using variables, arithmetic operators, and conditional statements.
- 4) Develop a program to calculate the factorial of a given number using variables, arithmetic operators, and loops.
- 5) Create a program to check if a given string is a palindrome or not using variables, string operators, and loops.

### **Array-related Problems:**

- 1) Java Program to copy all elements of one array into another array
- 2) Java Program to find the frequency of each element in the array
- 3) Java Program to left rotate the elements of an array
- 4) Java Program to print the duplicate elements of an array
- 5) Java Program to print the elements of an array
- 6) Java Program to print the elements of an array in reverse order
- 7) Java Program to print the elements of an array present on even position
- 8) Java Program to print the elements of an array present on odd position
- 9) Java Program to print the largest element in an array
- 10) Java Program to print the smallest element in an array
- 11) Java Program to print the number of elements present in an array
- 12) Java Program to print the sum of all the items of the array
- 13) Java Program to right rotate the elements of an array
- 14) Java Program to sort the elements of an array in ascending order

- 15) Java Program to sort the elements of an array in descending order
- 16) Find 3rd Largest Number in an Array
- 17) Find 2nd Largest Number in an Array
- 18) Find Largest Number in an Array
- 19) Find 2nd Smallest Number in an Array
- 20) Find Smallest Number in an Array
- 21) Remove Duplicate Element in an Array
- 22) Add Two Matrices
- 23) Multiply Two Matrices
- 24) Print Odd and Even Number from an Array